Unleashing UVM™

FIFO WITH WATERMARK SPECIFICATION

SystemVerilog, UVM - training, methodology consulting, auditing & debug

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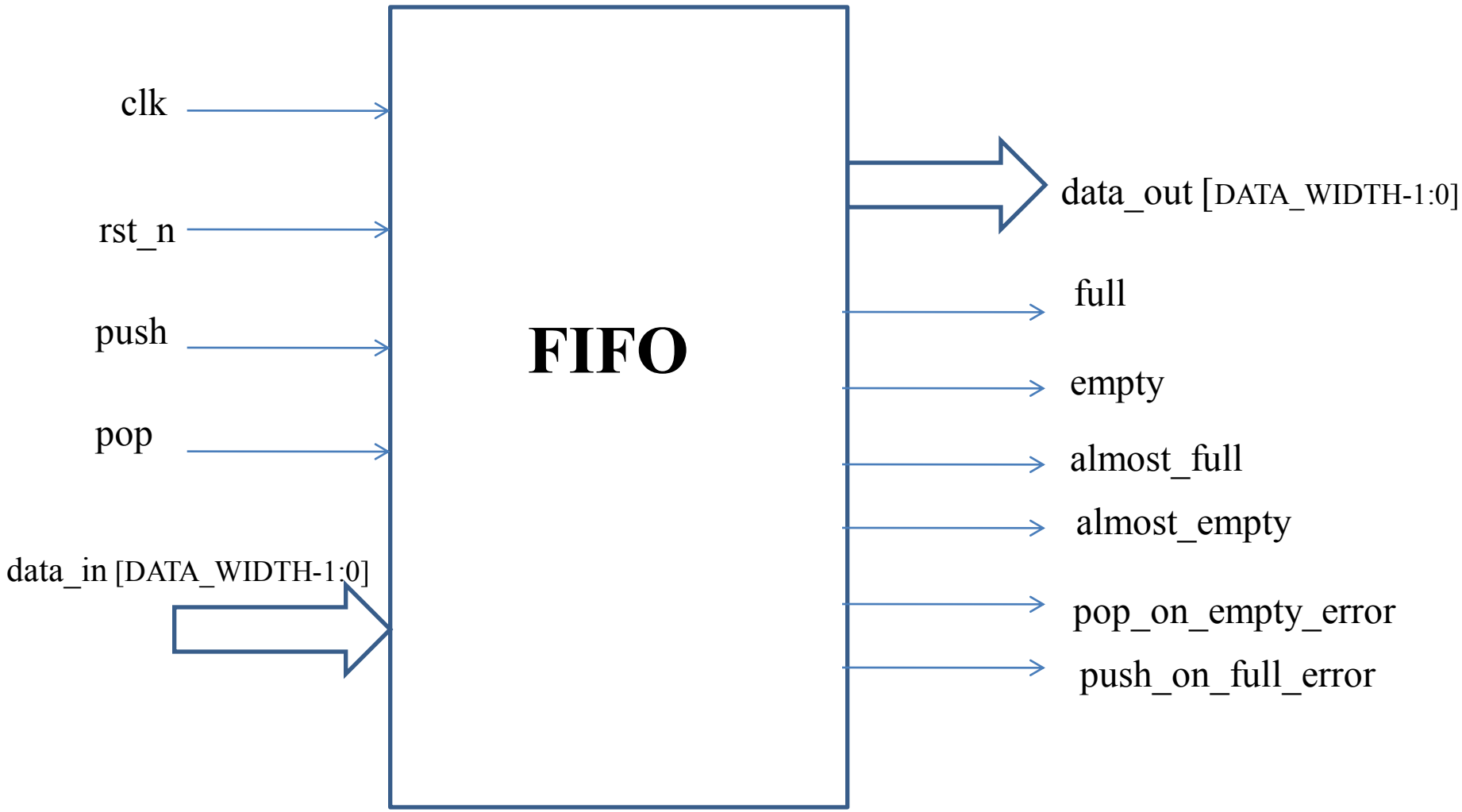
TOP MODULE :



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SIGNAL DESCRIPTION :

Signal	Width	Direction	Description
clk	1 bit	in	Synchronous clock for read and write (assumption 50 MHz)
rst_n	1 bit	in	Active low Reset
data_in	8 bits	in	Write data into the FIFO
push	1 bit	in	Push or write data into the FIFO, active high
pop	1 bit	in	Pop or read data from the FIFO, Active high
data_out	8 bits	Out	Read data from the FIFO
full	1 bit	Out	Indicates that the FIFO is full
empty	1 bit	Out	Indicates that the FIFO is empty
push_on_full_error	1 bit	Out	Indicates if trying to push data when FIFO is full
pop_on_Empty_error	1 bit	Out	Indicates if trying to pop data when FIFO is empty
almost_full	1 bit	Out	Indicates if FIFO reaches a upper limit
almost_empty	1 bit	Out	Indicates if FIFO reaches a lower limit

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CONTROL PATH DESIGN

FIFO depth=1K

- Parameters:
  - FIFO_DEPTH = 1K
  - DATA_WIDTH = 8
- PUSH & POP OPERATIONS
- PUSH
  - *Push signal* causes PUSH Operation.
  - Data into FIFO.
- POP
  - Pop signal causes POP Operation.
  - Data extracted from FIFO.

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REGISTERS:

Register Name	Width	Description
ALMOST_FULL_THRESHOLD	8 bits	If the FIFO level reaches the limit (FIFO_DEPTH - ALMOST_FULL_THRESHOLD), then the almost_full signal is High.
ALMOST_EMPTY_THRESHOLD	8 bits	If the FIFO level reaches this limit, then the almost_empty signal is High.

ALMOST_FULL_THRESHOLD = 10

ALMOST_EMPTY_THRESHOLD = 5

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Almost full/empty logic:

- Almost FULL

  As soon as fifo get filled beyond (FIFO_DEPTH – ALMOST_FULL_THRESHOLD), this signal should become HIGH

- Almost EMPTY

  As soon as fifo pops the data below(ALMOST_EMPTY_THRESHOLD), this signal should become HIGH

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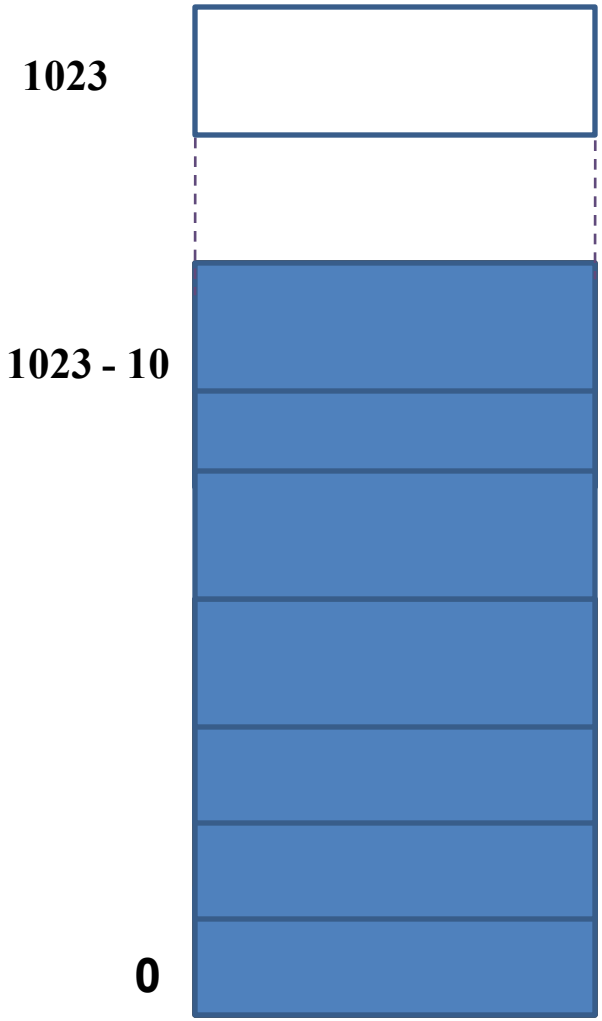
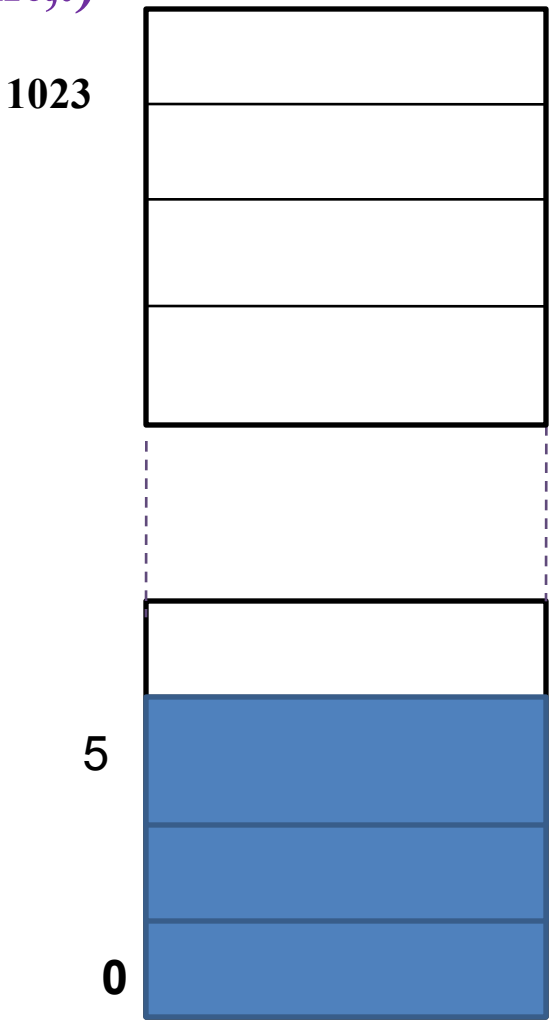
Almost full/empty logic (cont,.)



almost_full
signal becomes high



almost_empty signal
Becomes high.

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FIFO FULL Condition

- Let us take the FIFO depth for explanation purpose 4 and upper limit as 3

ACTION	DATA_WRITTEN
push	001
push	010
push	011
push	100

- If push is high for 3 locations(upper limit), then **almost_full** signal becomes high.
- If push is high for 4 locations, then **full** signal becomes high.
- If push is high for more than 4 Locations, then **push_on_full_error** signal becomes high and pushed data will not be stored in FIFO. Hence it's NOP.

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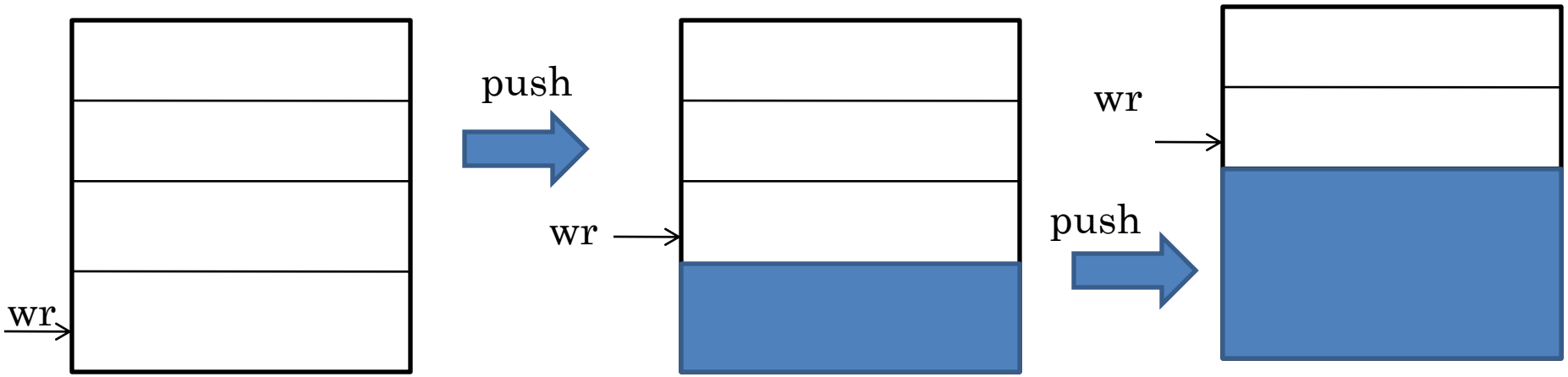
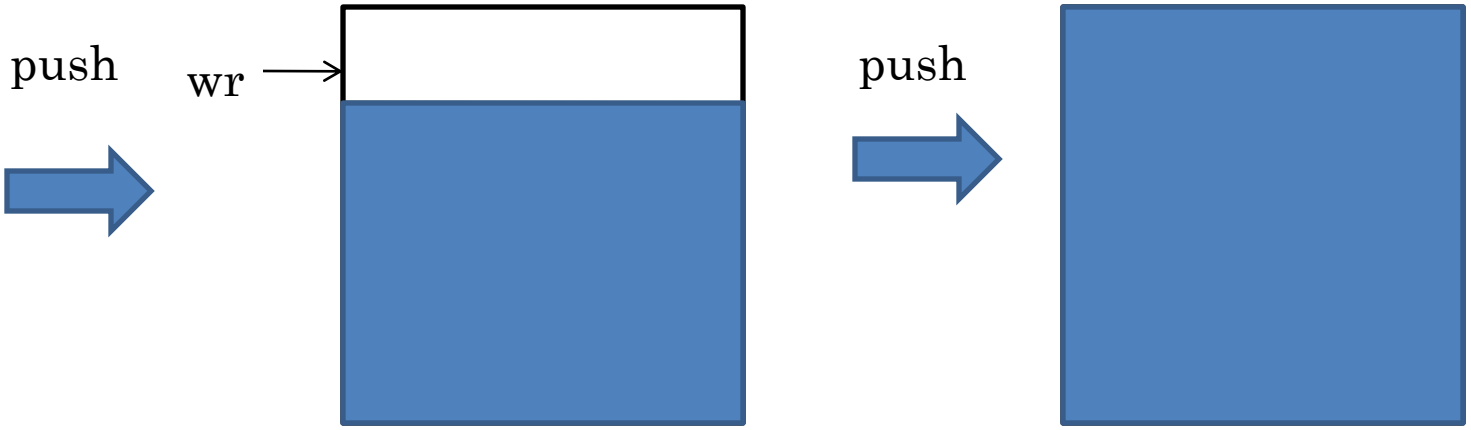
FLAG GENERATION :

push_on full_error signal



almost_full signal becomes high

If push is done more than FIFO memory.
push_on_full_error signal is High



full signal becomes high

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FIFO EMPTY CONDITION

- Let us take the FIFO depth for explanation purpose 4 and lower limit as 1.

ACTION	READ DATA
pop	001
pop	010
pop	011
pop	100

- If pop is high for 3 locations after 4 push operation, then **almost_empty** signal becomes high.
- If pop is high for 4 locations, then **empty** signal becomes high.
- If pop is high for more than 4 Locations, then **pop_on_empty_error** signal becomes high **.**

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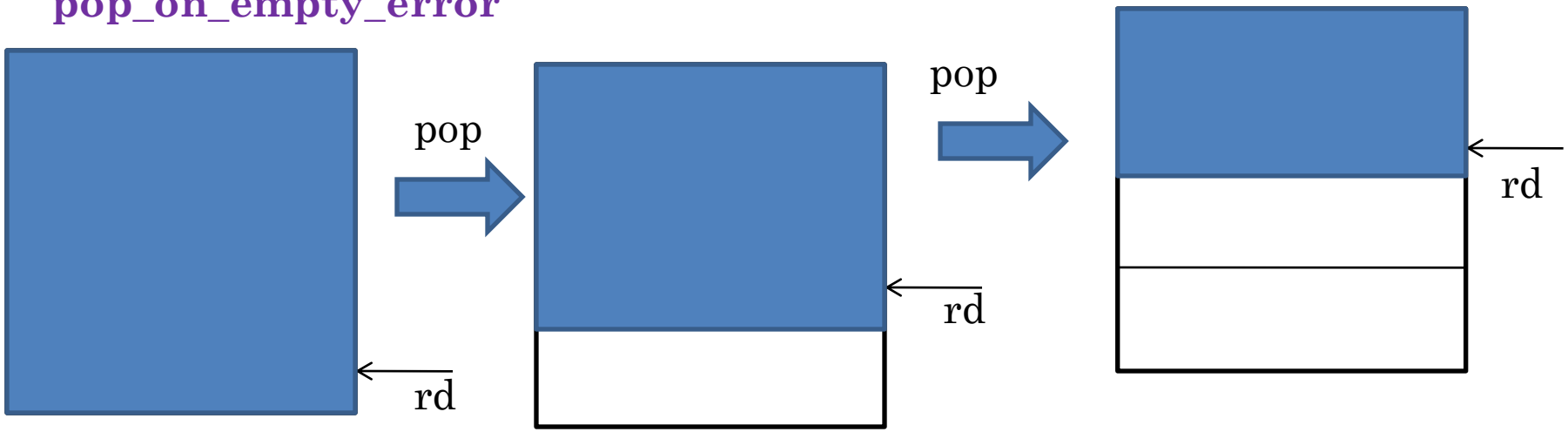
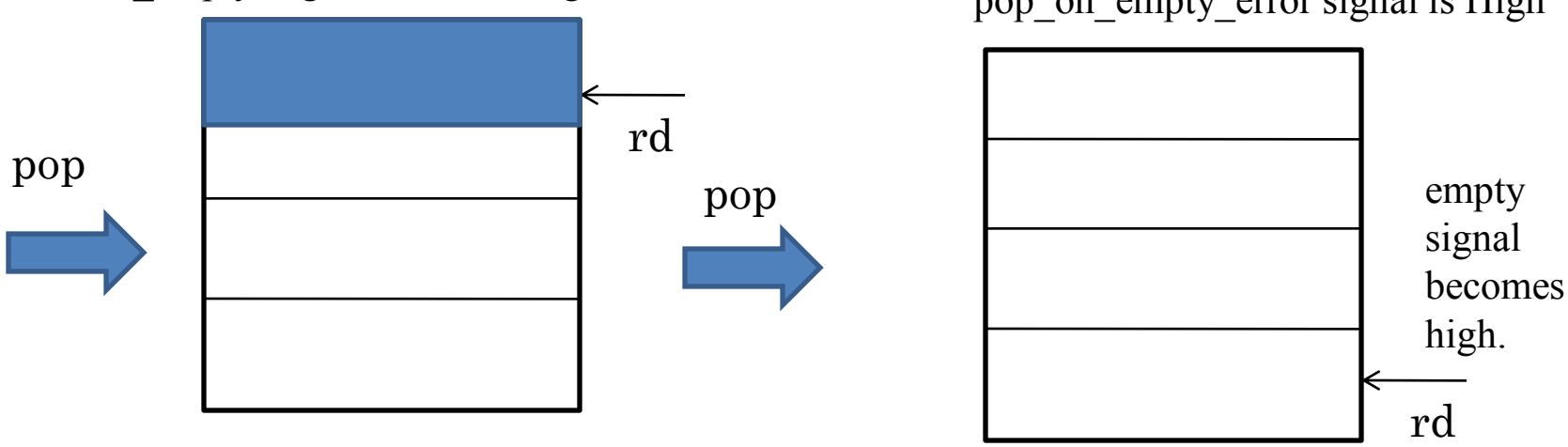
FLAG GENERATION :

pop_on_empty_error



If pop is done in unwritten FIFO memory pop_on_empty_error signal is High

almost_empty signal becomes high



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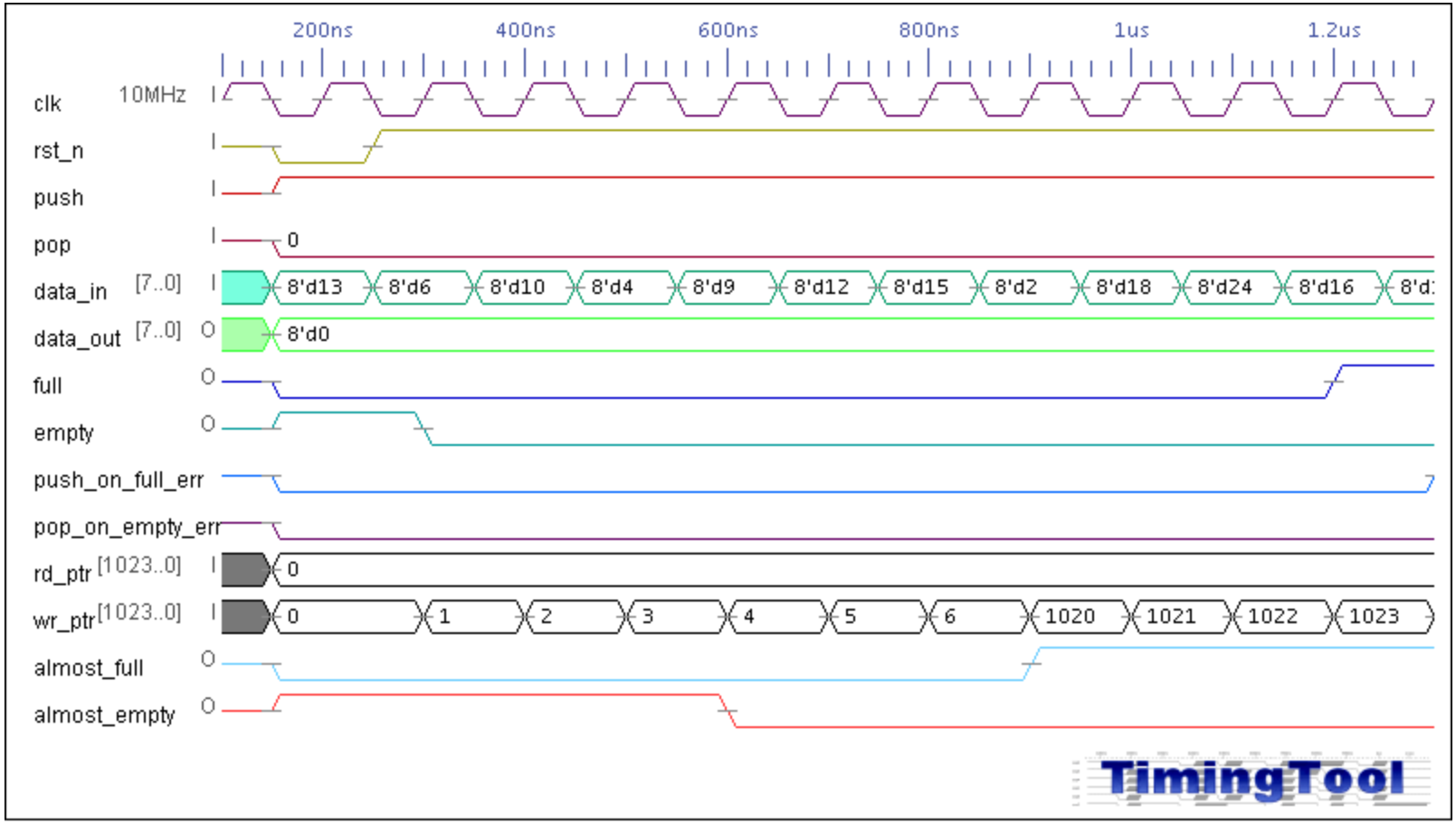
Timing_Diagram



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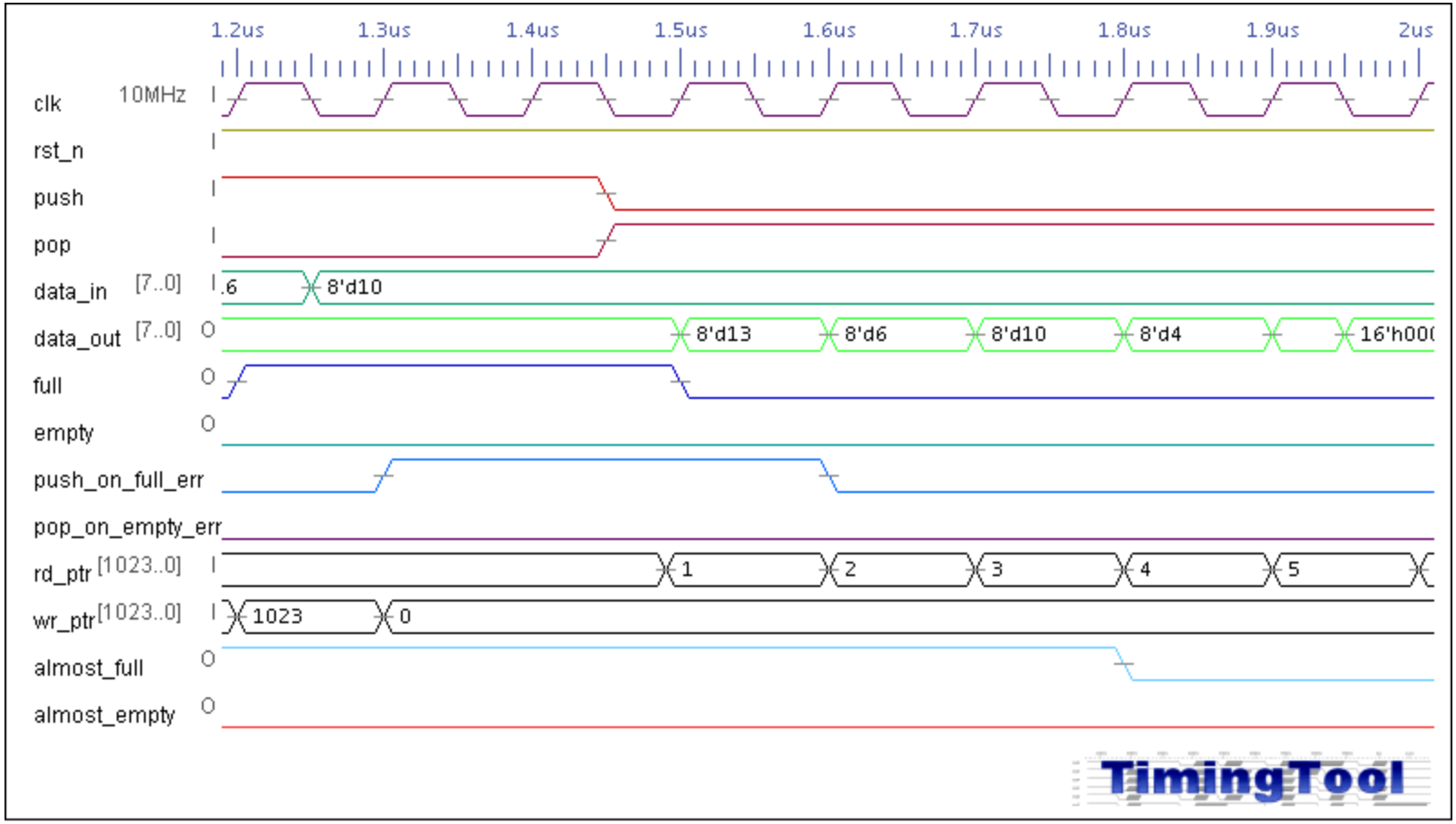

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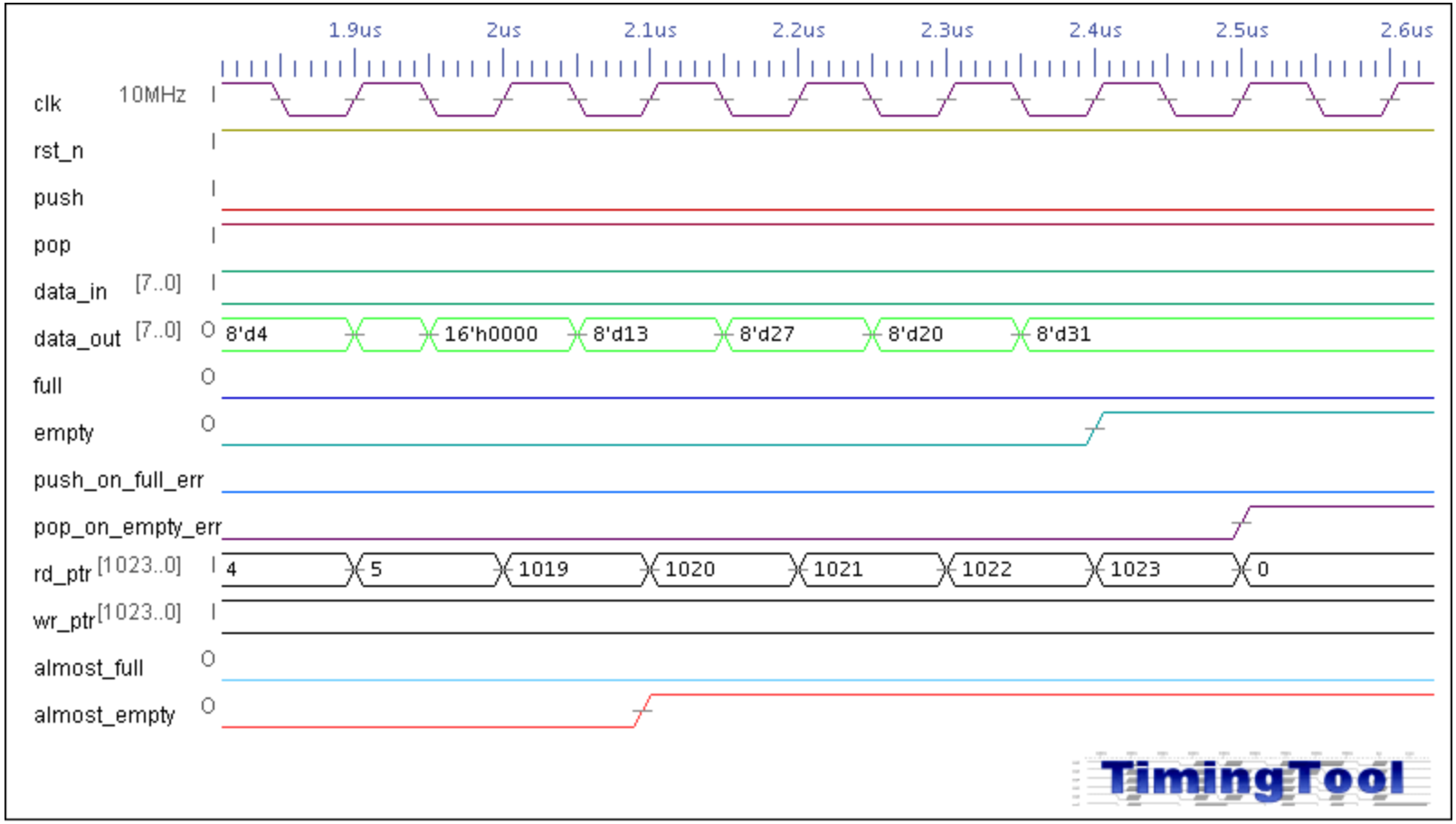

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